

STEADY

Treatment Response Fingerprint

Implementation-Ready Engineering Handoff

Clinical Intelligence Expansion | Document 02 | Repository-grounded against tprawlins-lang/Emdr | branch
claude/gifted-keller-501y5d | September 2026

Build directive

Build an individualized, evidence-linked model of how a person tends to respond to specific Steady and clinician-recorded interventions under specific conditions. The output describes observed response patterns, never proves causation, and becomes a reusable input to Session Prep, Recovery Trajectory, the Care Command Center, and Therapeutic Load.

Shared product rule

The clinician decides what clinical meaning and action to accept. Steady may measure, organize, compare, and suggest, but it does not silently turn an association, model output, or behavioral signal into clinical fact.

Repository foundation this handoff assumes

Foundation	Current Steady anchor	Rule for this build
Longitudinal history	src/lib/spine.ts + ADR 0010	Durable state must be event-backed and replayable. Do not create a parallel mutable history.
Tenant/person boundary	src/lib/repository.ts + ADR 0011	Every new durable patient record carries tenant_id and person_id; new paths use TenantContext even while legacy paths are migrating.
Clinician patient workspace	src/app/clinician/member/[id]/*	New clinician surfaces extend PersonShell and existing evidence/review conventions.
Evidence-linked summary	src/lib/clinical/summary.ts	A generated claim without resolvable evidence is withheld. Summary is a reading aid, not the record.
Trajectory	src/lib/clinical/trajectory.ts	Separate scales stay separate; provenance remains visible; do not invent a composite clinical meaning from arbitrary axis alignment.
Work queue	src/lib/clinical/work-queue.ts	Clinician home opens on actionable work, stable server ordering, clear reason, one primary action.
Companion memory	src/lib/companion.ts	Patient/Companion data retains source class and memory exposure rules. AI output is not clinician evidence.
AI Gateway	ADR 0012	New model tasks must route through the gateway with schema validation, task version, provenance, evaluation, and PHI policy.

This series also assumes the Clinician Thoughts / Clinical Memory / Session Prep Engineering Specification v2.1 is implemented or being implemented as the common clinician-memory and evidence layer.

Cross-feature invariants

- **Safety authority stays deterministic.** These features may add clinician decision support but cannot clear, weaken, bypass, or replace the safety engine.
- **Stabilization scope stays intact.** No feature autonomously initiates trauma reprocessing, changes a gated treatment decision, diagnoses, or issues a treatment order.
- **Source classes never collapse.** Patient report, clinician observation, system measurement, formal note, Companion interaction, and model inference remain distinguishable.
- **Associations are associations.** A temporal or statistical relationship is not described as causation unless a clinician explicitly records that judgement from appropriate evidence.
- **No future-data leakage.** Historical reconstruction and point-in-time evaluations use only information Steady knew at the relevant cutoff.
- **Missing data stays visible.** Missing observations do not become zero, normal, compliant, or recovered.
- **Corrections append.** An authorized correction supersedes prior derived state without erasing history.
- **Model output is disposable; evidence and provenance are permanent.** Every material intelligence output points to evidence IDs and task/policy versions.

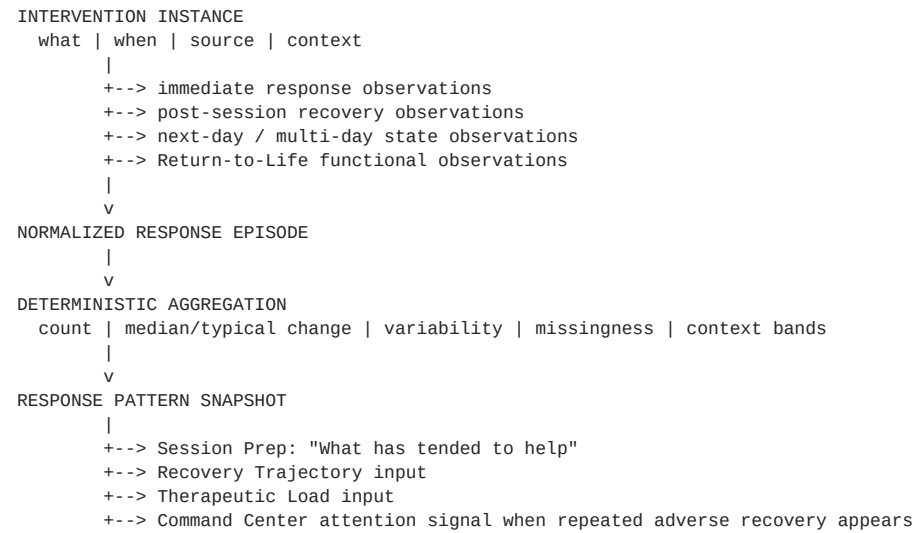
1. Product outcome

- Answer: "What has tended to help this person, under what circumstances, and what happens afterward?"
- Unify session response, grounding/practice use, Companion-supported interventions, clinician-recorded interventions, and functional outcomes into one intervention-response evidence model.
- Show counts, context, within-encounter change, delayed recovery, function change, and missing follow-up. Avoid a simplistic "effective / ineffective" label.
- Use Return-to-Life goal movement as one outcome channel, not the only outcome.

2. Intervention ontology

Class	Examples	Source
grounding	breathing, orienting, sensory grounding, walking	Existing safety plan, practices, clinician thoughts
resourcing	calm/safe place, containment, coping resource	Steady modules / clinician session
structured_practice	meditation, sleep practice, movement, lesson-linked exercise	Practice completion events
session_intervention	guided stabilization module, clinician intervention, BLS used within allowed scope	Therapy/session events + clinician documentation
companion_support	reflection, grounding prompt, planning conversation	Companion interactions; patient response required before claiming benefit
behavioral_action	return-to-life attempt, exposure-like real-world action only when clinician/patient records it	Goal observation / clinician note
external_clinician_entered	intervention outside Steady captured by clinician	Clinician Thoughts / formal note; requires normalized key

3. Data architecture



4. Database schema

```
CREATE TABLE intervention_definitions (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  canonical_key TEXT NOT NULL,
  display_name TEXT NOT NULL,
  intervention_class TEXT NOT NULL,
  source_scope TEXT NOT NULL,
  active INTEGER NOT NULL DEFAULT 1,
  created_at TEXT NOT NULL,
  UNIQUE(tenant_id, canonical_key)
);

CREATE TABLE intervention_instances (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
  intervention_definition_id TEXT NOT NULL,
  source_type TEXT NOT NULL,
  source_id TEXT NOT NULL,
  occurred_at TEXT NOT NULL,
  ended_at TEXT,
  dose_json TEXT NOT NULL DEFAULT '{}',
  context_json TEXT NOT NULL DEFAULT '{}',
  clinician_confirmed INTEGER NOT NULL DEFAULT 0,
  created_at TEXT NOT NULL
```

```

);

CREATE TABLE intervention_response_observations (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
  intervention_instance_id TEXT NOT NULL,
  outcome_type TEXT NOT NULL,
  window_type TEXT NOT NULL CHECK (window_type IN (
    'immediate', 'post_session', 'same_day', 'next_day', 'multi_day', 'functional'
  )),
  value_num REAL,
  value_text TEXT,
  unit TEXT,
  direction TEXT,
  evidence_class TEXT NOT NULL,
  source_type TEXT NOT NULL,
  source_id TEXT NOT NULL,
  occurred_at TEXT NOT NULL,
  created_at TEXT NOT NULL
);

CREATE TABLE response_fingerprint_snapshots (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
  intervention_definition_id TEXT NOT NULL,
  policy_version TEXT NOT NULL,
  evidence_cutoff TEXT NOT NULL,
  support_count INTEGER NOT NULL,
  missing_followup_count INTEGER NOT NULL,
  summary_json TEXT NOT NULL,
  limitations_json TEXT NOT NULL DEFAULT '[]',
  computed_at TEXT NOT NULL,
  UNIQUE(tenant_id, person_id, intervention_definition_id, policy_version, evidence_cutoff)
);

CREATE TABLE response_fingerprint_evidence (
  snapshot_id TEXT NOT NULL,
  evidence_type TEXT NOT NULL,
  evidence_id TEXT NOT NULL,
  PRIMARY KEY(snapshot_id, evidence_type, evidence_id)
);

```

5. Response dimensions

Dimension	Examples	Interpretation rule
within_encounter	pre/post SUDS, activation change	Describes immediate change only.
recovery_burden	hours elevated, post-session recovery confirmed, next-day distress	Delayed cost can coexist with immediate benefit.
sleep_after	same-night/next-night sleep quality	Association; do not call intervention causal.
function_after	Return-to-Life goal attempt or level observation	Highest-value real-life outcome when evidence is available.
patient_helpfulness	patient says helpful/not helpful/mixed	Patient report, not objective effect.
engagement_reuse	patient voluntarily reuses tool	Behavioral fact; not proof of benefit.
adverse_or_hard_stop	hard stop, worsening, inability to recover	Must remain visible and can create attention work.

6. Fingerprint computation

- Do not compute a pattern from one encounter. Default v1 display threshold: at least 3 comparable intervention instances, with a stronger "repeated pattern" label only after 5. Put thresholds in versioned policy.
- Use robust descriptive statistics: median change and observed range/IQR rather than a fragile mean when sample size is small.
- Separate context strata only when each stratum has enough support. Example: grounding during activation 5-7 versus activation 8-10.

- Missing delayed follow-up is reported. Do not classify it as recovered.
- An immediate distress decrease plus next-day worsening is displayed as mixed response, not netted into one number.

Fingerprint pattern state:
 insufficient_data
 mixed
 favorable_observed_pattern
 limited_observed_pattern
 recovery_burden_observed

Never use:
 "works"
 "effective treatment"
 "caused improvement"
 "contraindicated"
 without an independent clinician-authored judgement.

7. Events

Event	Meaning
intervention.instance_recorded.v1	One intervention exposure/instance occurred.
intervention.response_observed.v1	One source-backed outcome observation attached.
response_fingerprint.snapshot_computed.v1	Versioned deterministic pattern snapshot created.
response_fingerprint.pattern_reviewed.v1	Clinician reviewed a displayed pattern.
response_fingerprint.pattern_corrected.v1	Normalization/source mapping corrected without erasing prior history.

8. AI role

Task	Allowed	Not allowed
response.normalize_intervention	Map clinician wording to candidate canonical intervention	Auto-create clinical intervention identity without review when ambiguous.
response.summarize_pattern	Render deterministic stats into concise language with evidence IDs	Recalculate stats, invent causal explanation, suppress adverse observations.
response.match_context	Suggest thread/goal context related to an instance	Assert that context caused response.

9. UX

- **Patient overview:** "Observed responses" compact card with top 2-3 sufficiently supported interventions and a View all action.
- **Fingerprint detail:** each intervention gets support count, immediate response, delayed recovery, functional evidence, contexts, missing data, and source links.
- **Session Prep:** "What has tended to help / what to watch" using observed pattern language and evidence.
- **Clinician Thoughts:** an approved statement like "grounding worked well, 7 to 3" creates an intervention instance + response observation when the extraction is explicit enough.
- **Patient:** do not show clinical fingerprint labels by default. Patient-facing presentation should remain simple and non-evaluative if later exposed.

10. Integration adapters

Existing Steady source	Adapter behavior
therapy_sessions	Create session intervention instances and immediate pre/post response; keep incomplete/hard-stop rows visible.
practice_completions / intervention.completed events	Create practice instances; delayed response needs other evidence before any benefit pattern.

Existing Steady source	Adapter behavior
Companion	Create support instance only when a defined intervention occurred; model reply alone is not response evidence.
Return-to-Life observations	Attach functional response if temporally and semantically related; model match remains candidate.
Clinician Thoughts / formal notes	Can create explicit external intervention and patient/clinician response observations with source approval.
check-ins / measures	Supply before/after context windows, not direct causation.

11. Attention signal rules

- Handoff 03 will define the durable attention-signal contract. This handoff should expose a provider that can emit a non-safety signal when repeated recovery burden or repeated adverse response crosses a versioned threshold.
- Do not turn a single difficult session into a Command Center work item unless existing safety logic already does so.
- Do not duplicate existing hard-stop or urgent safety alerts as response-fingerprint signals.

12. Implementation phases

Phase 1 - Ontology + instances

Build

- Schema and canonical intervention registry
- Adapters for sessions and practices
- Manual clinician normalization

Definition of done

- Instances reconstruct from source events
- No benefit labels yet

Phase 2 - Response observations

Build

- Immediate + delayed + functional observation model
- Return-to-Life adapter
- Companion/check-in context adapters

Definition of done

- Mixed and missing outcomes remain visible

Phase 3 - Fingerprint projection

Build

- Versioned deterministic aggregator
- Clinician card/detail UI
- Session Prep adapter

Definition of done

- Minimum evidence thresholds enforced
- Every pattern opens evidence

Phase 4 - AI wording and downstream provider

Build

- Gateway summary task
- Non-safety attention signal provider interface for Handoff 03

Definition of done

- AI cannot change statistics or causal semantics

13. Acceptance criteria

- Session response keeps hard stops and missing closes visible, consistent with current Steady session-response design.
- Fingerprint never treats missing delayed data as recovery.
- Return-to-Life functional movement can appear as one response dimension.
- Patient report, clinician observation, and measured values retain provenance.
- No displayed pattern uses causal or efficacy language by default.
- All pattern summaries are reproducible from evidence + policy version.

Appendix A. File-level implementation map

Path	Change
src/lib/clinical/interventions.ts	NEW intervention ontology and instance adapters.
src/lib/clinical/response-fingerprint.ts	NEW aggregation/pattern projection.
src/lib/clinical/session-response.ts	EXTEND adapter, do not weaken missing-data semantics.
src/app/clinician/member/[id]/responses/page.tsx	NEW fingerprint detail view.
src/components/clinical/ResponseFingerprintCard.tsx	NEW patient overview/session-prep card.
src/lib/spine.ts + projections.ts	Add response events/projectors.
src/lib/db.ts + scripts/pg-schema.sql	Add/mirror schema.
src/lib/ai-gateway/*	Register normalization/summary tasks.
tests/response-fingerprint*.test.ts	Domain, mixed-response, missingness, security, replay, AI semantics.